

Kunyi (Mark) Ma

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Education

Columbia University , New York, NY Ph.D. Student in Mathematics, 4.15/4 Advisors: Daniela De Silva, Ovidiu Savin Research interest: PDE Analysis, Free Boundaries	2024–2029
New York University Shanghai , Shanghai, China B.S. in Honors Mathematics and Data Science, magna cum laude, 3.897/4 <i>Existence of Stationary Measure for Incompressible 2D Stochastic Navier-Stokes Equations</i> (with Vahagn Nersesyan) <i>On modifying Loss Function for PINO to approach Mathematics Nature in Fluid Mechanic</i> (with Zhuang Su)	2020–2024
Nanyang Model High School , Shanghai, China	2017–2020

Talks

Poster Presentations

Workshop on Harmonic Analysis and Partial Differential Equations, University of Pittsburg	Nov. 15th 2025
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Seminar Talks @ Columbia

Elliptic PDE Group Meeting

Caffarelli-Cabré: Jensen's Approximation and Uniqueness of viscosity solution to Dirichlet BVP	Feb. 11th 2026
Alt-Caffarelli: Viscosity Solution, Non-degeneracy, Lebesgue Density Estimate	Nov.7th 2025
Alt-Caffarelli: Euler-Lagrange Equations, Optimal Regularity	Nov.6th 2025
The Obstacle Problem: Singular Set, Monneau Monotonicity, ACF	Oct. 14th 2025
The Obstacle Problem: Classification of Blowups	Sep. 30th 2025
The Obstacle Problem: Blowups, 2-homogeneity, Convexity	Sep. 23nd 2025
The Obstacle Problem: Minimizers, EL equations, Continuity, Optimal Regularity, Nondegeneracy	Sep. 9th 2025

The Theory of Bounded Variation Functions & Regularity Analysis of the Minimal Surface PDE

De Giorgi's Lemma for C^1 Caccioppoli Sets	Nov. 6th 2024
De Giorgi's Lemma for Harmonic and C^1 function	Oct. 16th 2024
Regularity of the reduced boundary	Sep. 25th 2024
Basic properties of functions of bounded variation	Sep. 11th 2024

Informal Reading Seminar on GR

Local Theory for Quasi-linear Wave Equations	Nov. 18th 2024
Existence of Solutions to non-constant coefficient linear wave IVP	Oct. 7th 2024
Representation formula using Homogeneous Distribution	Sep. 16th 2024

Teachings

Teaching Assistant

Applied Functional Analysis APMA4150, Columbia University	Spring 2026
Ordinary Differential Equations UN2030, Columbia University	Spring 2026
Analysis & Probability I GR6151 (Graduate Level), Columbia University	Fall 2025
Linear Algebra S2010, Columbia University	Summer 2025 B
Analysis and Optimization S2500 (Grader), Columbia University	Summer 2025 B
Introduction to Modern Analysis I S4061, Columbia University	Summer 2025 A

Directed Reading Program

Introduction to Elliptic PDEs and Geometric Measure Theory (with Liam Dennehy, Marlet Ma)	Spring 2026
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Student Tutor

College Algebra and Calculus I–III, Columbia University	Fall 2024
ARC Learning Assistant for Pre-Calculus, NYU Shanghai	Fall 2021

Honors and Awards

NYU Shanghai

Magna cum laude; Phi Beta Kappa; NYU Shanghai Excellence Award; NYU Honors Scholar: Founders Day Award; 2023 NYU Shanghai Dean’s Undergraduate Research Funding; 2022 NYU Shanghai Hack the Pearl, Champion of Data Modelling Track; 2020–2022 NYU Shanghai Recognition Award; 2020–2023 NYU Shanghai Dean’s List

Attendance

SIAM Conference on Analysis of Partial Differential Equations (PD25)	Nov. 2025
Workshop on Harmonic Analysis and Partial Differential Equations, University of Pittsburg	Nov. 2025
The Third Joint SIAM/CAIMS Annual Meetings (AN25), Montréal, Québec, Canada	July 2025
PDE Summer School 2025 Theory and Applications of Elliptic PDE, UC Irvine	June 2025
Rocky Mountain Mathematics Consortium Summer School 2025, ASU, University of Wyoming	June 2025
Duke Mathematics Journal 2025 Conference, Duke University	Apr. 2025
Winter School: Boundary and Singularity in Fluid Mechanics, Simons Center	Jan. 2025
Differential Geometry Summer School 2024, PKU BICMR	July-Aug. 2024
Workshop on Minimal Surfaces and Mean Curvature Flow 2024, PKU BICMR	July 2024
International Congress of Chinese Mathematicians 2022, THU YMSC	July-Aug. 2022

Extracurricular

Orientation Ambassador, NYU Shanghai	May–Sep. 2021
President, Literature@NYUSH Student Club, NYU Shanghai	Jun. 2021–May 2022
Tenor, NYU Shanghai Elite A Cappella Chamber Singers	Feb. 2021–Dec. 2022
Quantitative Researcher, Ipsos China, Shanghai	Jun.–Aug. 2021